Ivan Turik

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Technical Skills

Python, SQL, C++

pandas, NumPy, scikit-learn, imbalanced-learn, CatBoost, PyTorch, matplotlib

FastAPI, Streamlit, MLflow, Docker, Git, MCP, Hugging Face Transformers, A/B testing

Projects

- **Heart Disease Scoring — End-to-End ML Service**
  - Developed a pipeline to retrieve source data from **CDC NHANES** across **5 survey cycles**; merged heterogeneous XPT tables, harmonized variable coding and assembled **28,270 labeled records**.
  - Performed **EDA** of feature distributions and missingness; built preprocessing and train/validation/test splits, grouping duplicate feature profiles to reduce leakage.
  - Compared **8 models** and tracked experiments in **MLflow**; selected **CatBoost** by validation Average Precision and tuned the threshold using MCC with a recall constraint. Test: **Recall 0.752**, Precision 0.219.
  - Turned the model into a usable app: **FastAPI** with input validation and missing-value handling, a **Streamlit UI** with clear result explanations, and **Docker Compose** for local deployment.
- **DocQA — Extractive Question Answering**
  - Fine-tuned **DistilBERT** on **SQuAD 1.1** for extractive question answering; implemented sliding-window processing for long contexts and optimized training with mixed precision.
  - Achieved **76.93% Exact Match** and **85.18% token-level F1** on a held-out set; analyzed answer-span boundary errors and performance across context lengths.
  - Integrated the fine-tuned model into a **Streamlit application** for interactive testing and result visualization.
- **Study of Methods for Handling Imbalanced Data in Binary Classification**
  - Benchmarked **15 approaches** to class imbalance (SMOTE, class weighting, ensembles and threshold tuning) across **4 real-world datasets** and synthetic scenarios with **10:1–1000:1** class ratios.
  - Studied how noise, class overlap and resampling intensity affect performance and training cost. SMOTE-based methods with Random Forest generally offered the best precision/recall balance; achieved **mean F1 = 0.866** on Fraud Detection over 3 splits.
  - On Fraud Detection, demonstrated how SMOTE + RF increased **recall from 0.770 to 0.831** versus Random Oversampling + RF at comparable F1 but caused more false alarms; with **1000:1** imbalance, partial balancing outperformed a 1:1 class ratio.

Education and Achievements

Belarusian State University, Faculty of Applied Mathematics and Computer Science *2023–2027*

Applied Mathematics (expected graduation 2027)

Courses: Yandex School of Data Analysis Agents Week; "Training: ML Race" (RL theory); BSU FAMCS Student Union ML course (Season 1).

Olympiads & Competitions: Final stage of the Belarusian Republican Mathematics Olympiad (2023); World Robot Olympiad (WRO) International Final, Open Elementary category (2016, 2017).

Languages

Russian — native • English — B2